

Singular support and characteristic cycle

Recall. What's SS of ① a distribution on (real) mfd?

② a (bounded) complex of constructible sheaves on (real) mfd?

$$SS(F) := \{ (x, \zeta) \in T^*X \mid \begin{array}{l} \text{conic} \\ \forall U \ni (x, \zeta), \forall \psi \in C_c^\infty(U), \psi(x) \neq 0 \\ \text{then } \int_U \psi(x) (1+i\zeta)^N \rightarrow \infty \text{ when } |z| \rightarrow \infty, \exists N \end{array} \}$$

(called wavefront than singular support, in analysis setting, SS is in X)

$$SS(F) := \{ (x, \zeta) \in T^*X \mid \begin{array}{l} \text{conic} \\ \forall U \ni (x, \zeta), \forall \psi \in C_c^\infty(U), \psi(x) \neq 0 \\ \text{then } (R\Gamma_{\{x\}}(F))_x \neq 0 \end{array} \}$$

they're in the form: $\begin{array}{l} \text{Morse function } (\pi(U), \partial \phi(x(U))) \\ f(x, \zeta) \in T^*X \mid \forall U \ni (x, \zeta), \text{ test on } \pi(U), \text{ then not microlocally smooth / locally constant} \end{array}$

note that there is only two ingredients: $\pi(U)$ and ψ , they

recover the U , i.e. for any $V \ni x$ and $\forall \psi \in C_c^\infty(V)$ enough

We can rewrite them into such form, outlining the V and ψ :

$$SS = \bigcap_{C \in \mathcal{C}} C, \quad \mathcal{C} = \{ C \subset T^*X \text{ conic} \mid \begin{array}{l} \forall (V, \psi) \text{ s.t. } \partial \psi(V) \subset C \\ \text{then } \square \text{ holds} \end{array} \}$$

Ambiguity. ① Why $\bigcap C$ is still closed?

② Why $\bigcap C$ is still conic?

Both are trivial here, but far-reach in next generalization.

PART I Bellison's singular support in arbitrary characteristic

Our key idea is to generalize the data (V, ψ) on mfd.

Now let X is smooth scheme / k , our data (V, ψ) should be

naturally refined by (f, g) , where f is étale and g is

"function", that is: $\mathbb{A}^1 \xrightarrow{f} X$ where g étale determines

the " $V \subset X$ " above in $\mathbb{A}^1$ algebraic setting for constructibility.



日月光华 旦复旦兮

Now the technical issue is: how to define $\mathcal{C}$? We need a relative version to pullback C along f . then let $(dg)^{-1}(\mathbb{A}^1) \subset \square$, this

is the notation of C -transversality (Here C is cone over X)

then we define $SS(F) = \bigcap_{C \in \mathcal{C}} C$, where $\mathcal{C} = \{ C \subset T^*X \text{ conic} \mid$

is called admissible set. $\mathcal{C} \in \mathcal{C} \iff \forall Z \xrightarrow{f} X, (f, g) \text{ is } C\text{-transversal}$

③ The singular support is the place avoiding some "good condition" microlocally. Here our nearby cycle

vanishes étale locally on X is called g is locally

acyclic relatively to f^*F , which is the relative version of microlocally

constancy. The constancy and smoothness are equivalent under

Brothendick's sheaf-to-function dictionary: the graph of smooth

function is diffeomorphic to constant function $\bigwedge \cong \text{---}$ locally

and can be realized as section of locally constant sheaf, thus either

the propagation of singularity, or the propagation of support, they're same.

E.g. $WF(S_Y) = N_Y^*X = SS(\underline{S}_Y)$ for $Y \subset X$ closed.

C-transversality

Recall the classical transversality of a smooth map and a

submfd: $f: Y \rightarrow X \supset Z: i$, (f, i) is transversal $\iff$

$\forall p \in Z, \forall q \in f^{-1}(p) \subset Y, (f_*)_q T_q Y \cap N_p^*Z = \{0\}$, where

$(f_*)_q: T_q Y \rightarrow T_p X$, globally it's $T_Y \xrightarrow{f_*} f^*T_X \supset f^*T_Z = (f^*T_Z)_q$

Now we replace $T_p Z$ by cone, first $\downarrow$ we pass to its dual

$\iff \forall p \in Z, \forall q \in f^{-1}(p) \subset Y, (f^*)_q T_q Y \cap N_p^*Z = \{0\}$

$(f^*)_q T_q Y = (f^*)_q(T_q Y)$, where $T_Y \xleftarrow{f^*} f^*T_X$

replace $T_p Z$ by C $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$

$f^*C = (f^*C)_q$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$

$f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$

$f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$

$f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$

$f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$ $\iff$ $f^*C \cap (f^*)_q(T_q Y) = \{0\}$

Def. ① $f: Y \rightarrow X$, X, Y both smooth (but f needn't), $C \subset T^*X$, f is called C -transversal iff $f^*C \cap (df)^*T^*Y = \emptyset$;

② $C \subset T^*Y$, f is called C -transversal iff $(df)^*C \subset \text{so}(Y)$;

Rk ① is motivated from transversality; ② is motivated from our need, or the transversality of $i: Z \subset Y \xrightarrow{f} X$;

Prop. ① In the first case, it's equivalent to say $f^*C \cap (df)^*\text{so}(Y) = \emptyset$ where $f^*T^*X \xrightarrow{df} T^*Y$;
 $\begin{array}{ccc} \text{so}(X) & \hookrightarrow & T^*X \\ & \searrow & \swarrow \\ & Y & \hookrightarrow & X \end{array}$

② In the first case, it's equivalent to say $(df)|_{f^*C}: f^*C \rightarrow (df)^*\text{so}(Y)$ is finite, we say its image $f^*C \subset T^*Y$;

③ If f smooth in the first case, $(df)|_{f^*C}$ is isomorphism;

④ In both case, the C -transversal loci is open. Here the C -transversality at each pt is defined fibrewise.

Pr. ① Trivial, only $\emptyset$ can be pulled back to zero;

② $(df)^*\text{so}(Y)$ go out of $\text{so}(Y)$ at degeneracy locus of f ;

③ By ②;

④ $C \subset \text{so}(Y) \Leftrightarrow f^*C = \emptyset$, thus the complement of C -transversal loci is the projection of $P(f^*C \cap (df)^*T^*Y)$, which is closed $\square$

Def. $\mathbb{A}^1 \times Y \xrightarrow{g} Y \xrightarrow{f} X$, (f, g) is C -transversal pair, where $C \subset T^*X$ iff
 • f is C -transversal
 • g is f^*C -transversal

Prop. ① (f, g) is C -transversal pair $\Leftrightarrow (f, g)$ is $C \times T^*Z$ -transversal as $(f, g): X \rightarrow Y \times Z \leftarrow T^*Y \times T^*Z \supset C \times T^*Z$

Pr. Definition $\square$



日月光华 旦复旦兮

Rk. Besides the smooth pullback f^*C , we also have proper pushforward f_*C :

$$\begin{array}{ccccc} T^*X & \xleftarrow{f^*} & T^*Y & \xleftarrow{g^*} & T^*Z \\ \cup & & \cup & & \cup \\ f_*C & \xleftarrow{f_*} & f^*C & \xleftarrow{g^*} & C \end{array} \quad \begin{array}{c} \text{proper pushforward} \\ \text{smooth (flat) pullback} \end{array}$$

PART III Characteristic cycle and Milnor formula

Now we have $\text{SS}(F) \subset T^*X$ closed, conic, isotropic

thus $\text{SS}(F) = \bigcup C_i$ decompose it into irreducible components

$$C_i = N_{T^*X}^{\perp} \subset T^*X$$

We set the characteristic cycle $\text{CC}(F) \in H_*(T^*X)$ (or Chow)

$\text{CC}(F) := \sum m_i [C_i]$, the microlocal multiplicity m_i are determ

by Milnor formula: $\chi(\mathbb{E}_F)_x = \langle \text{CC}(F), \mathbb{I}_x \rangle_{\text{local}}$
 Global topological local intersection number

Fact.

① Those m_i do coincide with our definition in CSM classes, we'll see why it's a result of graph construction. Indeed, $\begin{array}{c} T^*X \rightarrow X \\ \text{CC} \xrightarrow{\text{graph}} \text{CSM} \end{array}$; see the historical survey next;

② Technically, it's ~~not~~ proven by embedding $\mathbb{A}^1$ to projective space, then pullback, see [Saito, Characteristic cycle of constructible sheaf]. Historically, Milnor studied Milnor fibration $f: (C^{n+1}, 0) \rightarrow (C^1, 0)$ with $\frac{\partial f}{\partial x_i}|_{x=0}$ degenerate and $\frac{\partial f}{\partial x_i}|_{x \neq 0}$ non-degenerate.

the Milnor number: $\mu := \dim_C (C[x_0, \dots, x_n] / (\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}))$

the Milnor formulae: $\langle \mathbb{E}_F - 1, (-1)^n \rangle = \mu$
 general fiber shift from perversity $\Rightarrow$ this is origin of our Milnor formula above.

the intersection number: $\mu = \langle \mathbb{I}_0, \text{CC}(F) \rangle_{\text{local}}$

$$\begin{array}{l} \text{CC}(\mathbb{C}) = \text{zero section} \\ \text{or (characteristic) fiber of } \pi \end{array}$$

日月光华 旦复旦兮

In our construction of CSM class, we consider relative $X \xrightarrow{f} Y$, let $Y = \text{pt}$, then $\text{Gr}(d, TX \oplus TY) = \text{Gr}(d, TX)$, and the graph is $\bullet$ the single fibre. We have Γ_{α_i} with multiplicity m_i , but intersect with $[\hat{V}_i]$ with multiplicity k_i , then our desired microlocal multiplicity is $m_i k_i =: n_i$ on Y . When we do the Segre class operation, the microlocal multiplicity $\underline{m}_i$ on T^*X will multiply ~~to~~ the contribution of "zero section" k_i , push to $\underline{n}_i$ on X , which is different, take care of it!

Rk. A surprising result is: we denote $\text{CC}_C(F^*) = \sum m_i [C_i]$, $C = \cup C_i$, and m_i is determined by Milnor formulae. But the local data of $\mathbb{E}$ not depend on C , thus $\text{CC}(F^*) := \text{CC}_C(F^*)$ for $\forall C$ is well-defined. Then we can remove our definition of $\text{SS}(F)$, i.e. without the need of a minimal element in admissible cones. Thus for all $\mathbb{K}$ perfect, we can define CC , while defining SS acquires stricter finiteness.